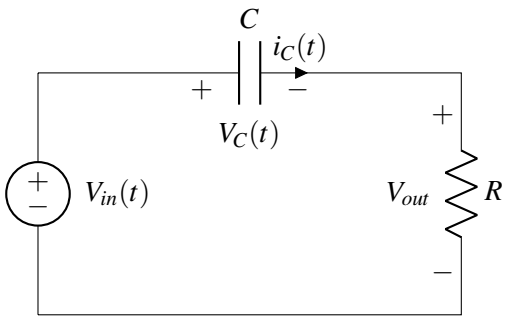
CSM
Fall 2025

EECS 16B
Worksheet 5

1. CR Circuit

We're already familiar with the charging RC circuit from the second problem of this worksheet, but what happens if we flip the orientation of the circuit components and switch to discharging?

Consider the CR circuit below:



(a) **Write out the differential equation for the voltage $V_C(t)$ across the capacitor in terms of constants and $V_{in}(t)$.**

(b) Assume that when $t \leq 0$, the capacitor has been fully charged with an input voltage V_{DD}, with the initial condition $V_C(t=0) = V_{DD}$. At $t = 0$, the input voltage switches from high to low, so that $V_{in}(t) = 0$ for $t \geq 0$.

Plug in these conditions to the differential equation from the previous part and solve for $V_C(t)$ for $t \geq 0$.

(c) **What is $V_{out}(t)$?** Sketch a plot of the voltage across the resistor over time, labeling the asymptote it reaches at steady-state.

(d) **What is the steady-state voltage across the capacitor as $t \to \infty$?**

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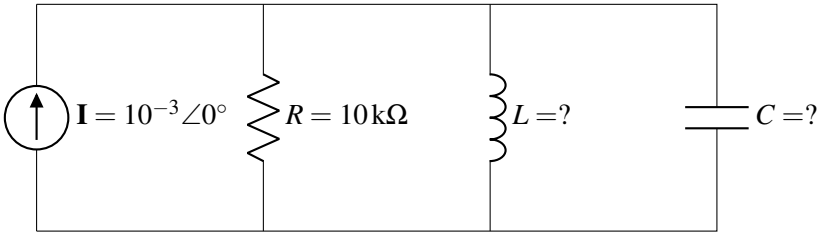
(e) **What is the steady-state current across the capacitor as $t \to \infty$?**

(f) **What circuit element does the capacitor act like at steady-state ($t \to \infty$)?**

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2. Parallel Resonant Circuit

(a) Find the L and C values for a parallel resonant circuit that has $R = 10\,\text{k}\Omega$, $f_0 = 1\,\text{MHz}$, and $B = 100\,\text{kHz}$. You are also told that $\mathbf{I} = 10^{-3}\angle 0^\circ$ A.



(b) Calculate the current phasors through each of the elements at resonance.

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3. Fun with Inductors (Challenge)

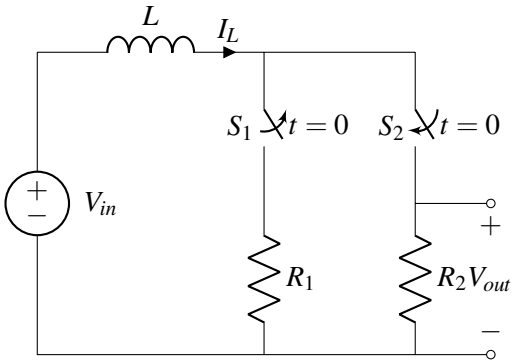


Figure 1: Circuit A

(a) Consider circuit A. Assuming that for $t < 0$, switch S_1 is on and switch S_2 is off (and both switches have been in these states indefinitely), what is $i_L(0)$?

(b) Now let's assume that for $t \geq 0$, S_1 is off and S_2 is on. Solve for $V_{out}(t)$ for $t \geq 0$.

(c) If $V_{in} = 1V$, $L = 1nH$, $R_1 = 1k\Omega$, and $R_2 = 10k\Omega$, what is the maximum value of $V_{out}(t)$ for $t \geq 0$?

(d) In general, if we want $\max V_{out}(t)$ to be greater than V_{in}, what relationship needs to be maintained between the values of R_1 and R_2?

(e) Now assume that at time $t = t_1$, switch S_2 was turned off, and switch S_1 was turned back on. Solve for $i_L(t)$ for $t > t_1$. If $R_2 > R_1$, how does this $i_L(t)$ for $t > t_1$ compare with the initial condition $i_L(0)$ you found in part (a)?

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